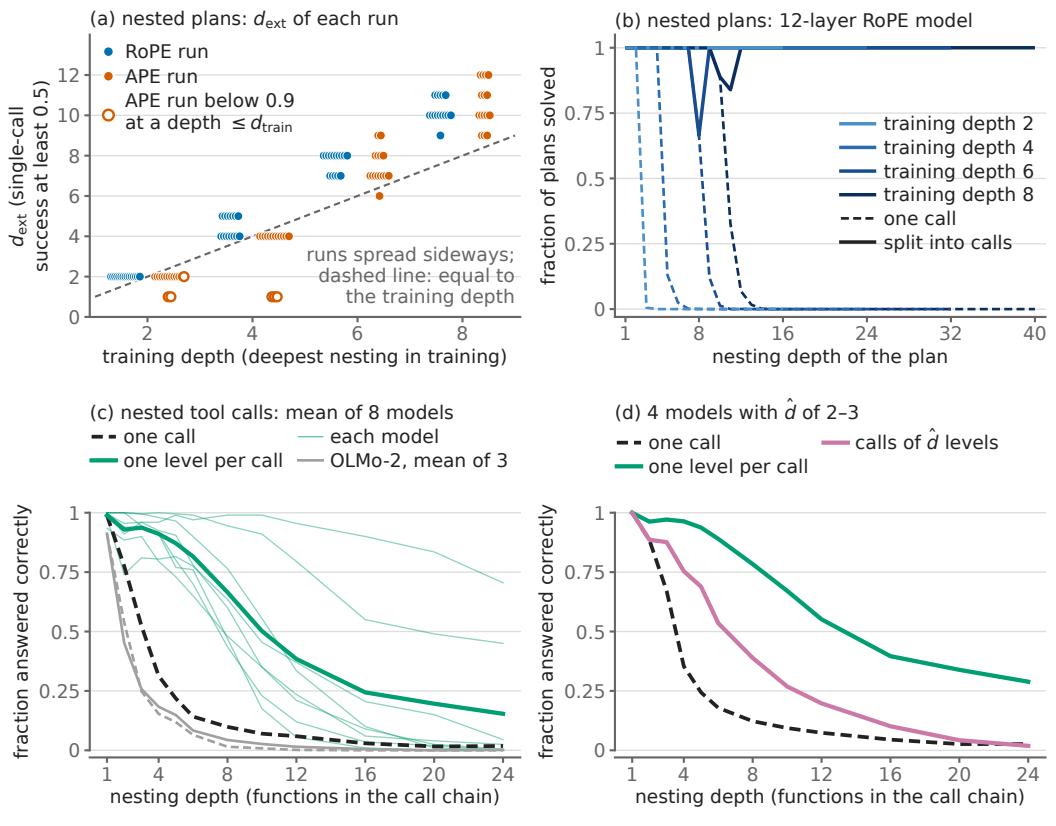
(a) nested plans: d_{ext} of each run
RoPE run
APE run
APE run below 0.9 at a depth $\leq d_{train}$
runs spread sideways; dashed line: equal to the training depth
d_{ext} (single-call success at least 0.5)
training depth (deepest nesting in training)
(b) nested plans: 12-layer RoPE model
training depth 2
training depth 4
training depth 6
training depth 8
one call
split into calls
fraction of plans solved
nesting depth of the plan
(c) nested tool calls: mean of 8 models
one call
one level per call
each model
OLMo-2, mean of 3
fraction answered correctly
nesting depth (functions in the call chain)
(d) 4 models with $\hat{d}$ of 2–3
one call
one level per call
calls of $\hat{d}$ levels
fraction answered correctly
nesting depth (functions in the call chain)